

QUICKFOX FDMTP DLL SIDELOADING

Campaign hunt for csmonitor.exe loading a co-located Microsoft.ServiceHosting.Tools.dll from the source-observed QuickFox staging directory.

HUNTING QUERY

CONFIDENCE: MEDIUM

SOURCE-CONFIRMED BEHAVIOR

retain facts, isolate interpretation

DeviceImageLoadEvents

direct telemetry boundary

HUNT · VALIDATE · TUNE

no automatic malicious verdict

REPORT DATE

05 August 2026

VERSION

v1.0

PRIMARY TELEMETRY

Microsoft Defender XDR ·
DeviceImageLoadEvents

ATT&CK

T1574.001

Evidence-led dossier. Query status remains an untested implementation sketch until executed in the target environment.

INDEPENDENT RESEARCH FORMAT · CONTROLLED UI/UX MASTER



EXECUTIVE ASSESSMENT

Decision-grade summary for detection engineering and SOC operations.

ASSESSMENT

EXECUTIVE SUMMARY

Hunt the complete loader-module-path relationship.

FortiGuard Incident Response observed both FDMTP loader generations executing legitimate csmonitor.exe to side-load a malicious Microsoft.ServiceHosting.Tools.dll from the same quickfox\updated directory. The hunt retains all three source-observed elements.

OPERATIONAL VALUE

A precise endpoint pivot without relying on an IOC list.

The process-module relationship is directly observable in DLL-load telemetry and provides file hashes, paths and device context for scoping. It remains campaign-specific because the filenames and staging directory are mutable.

8.5

01

OPERATIONAL VALUE



Medium

02

DETECTION CONFIDENCE



Hunting

03

OPERATIONAL READINESS



SOURCE-CONFIRMED FACTS

Observed behavior

- Both loader generations use legitimate csmonitor.exe.
- The co-located Microsoft.ServiceHosting.Tools.dll is malicious in the analyzed chain.
- The archive is extracted under a quickfox\updated directory before csmonitor.exe is executed.

ANALYTICAL ASSESSMENT

What the evidence supports

- The complete relationship is substantially stronger than either filename alone.
- Same-directory loading supports the side-loading hypothesis without inspecting file contents.
- A match is a campaign hunt lead, not proof of FDMTP execution or attribution.

DETECTION ASSUMPTIONS

What must be validated

- DeviceImageLoadEvents records the DLL load in the tenant.
- FolderPath and InitiatingProcessFolderPath are populated and comparable.
- The source-observed directory suffix remains unchanged.

Confidence boundary

The exact filenames and directory are campaign artifacts. Legitimate Azure emulator components or security testing may overlap, and the event does not expose the DLL contents.

02

THREAT MECHANICS

Source-observed chain and explicit observability boundaries.

ATTACK FLOW

01

Trojanized QuickFox

Modified Electron renderer HTML downloads JavaScript.

02

Target validation

The script checks running processes and exits on guardrail failures.

03

Stage archive

update.zip is written under a QuickFox temporary path.

04

Extract loader set

csmonitor.exe and the malicious DLL are co-located under quickfox\updated.

05

Side-load payload

csmonitor.exe loads Microsoft.ServiceHosting.Tools.dll and starts the FDMTP chain.

Observable core

One DLL-load event tying csmonitor.exe to a co-located Microsoft.ServiceHosting.Tools.dll under a directory ending in quickfox\updated.

Non-observable boundary

Trojanized HTML content, JavaScript execution body, successful DLL payload behavior, C2 response, plugin execution and attribution.

DETECTION HYPOTHESIS

The source-observed side-load should create a precise module event.

If an attacker repeats the documented QuickFox loader chain, DeviceImageLoadEvents should show csmonitor.exe loading Microsoft.ServiceHosting.Tools.dll from the same quickfox\updated directory. Legitimate activity can be separated through file hashes, signatures, origin, software inventory and follow-on process or network evidence.

REQUIRED TELEMETRY

DeviceImageLoadEvents

- Microsoft Defender for Endpoint DLL-load events.
- Loaded module and initiating-process names and paths.
- Device and unique event identifiers.

OPTIONAL ENRICHMENT

Context that improves disposition

- SHA1, signer and prevalence for both files.
- DeviceFileEvents for archive extraction and data.dat creation.
- DeviceProcessEvents and DeviceNetworkEvents for follow-on activity.

UNAVAILABLE HERE

Do not infer

- HTML or JavaScript body from DLL-load telemetry.
- Successful FDMTP C2 or plugin execution.
- Campaign attribution from this event alone.



TELEMETRY & EVIDENCE MODEL

Minimum data contract and quality gates for a defensible hunt.

DATA CONTRACT

PRIMARY DATA SOURCE

Microsoft Defender XDR - DeviceImageLoadEvents

Contains DLL-load events with loaded module, initiating process, paths, hashes, account, device and event identifiers.

EVIDENCE BOUNDARY

A matching event is a lead, not a verdict.

The table can establish that a module was loaded by a process. It cannot establish that the loaded bytes are malicious without file inspection or corroborating behavior.

REQUIRED AND ENRICHMENT FIELD MODEL

CATEGORY	FIELD	PURPOSE	REQUIREMENT
Module event	DeviceImageLoadEvents	Primary DLL-load evidence.	REQUIRED
Loaded module	FileName, FolderPath	Module identity, location and suffix filter.	REQUIRED
Loader	InitiatingProcessFileName	csmonitor.exe identity.	REQUIRED
Loader path	InitiatingProcessFolderPath	Same-directory comparison.	REQUIRED
Device	DeviceId, DeviceName	Endpoint pivot.	REQUIRED
Event identity	Timestamp, ReportId	Unique-event investigation.	REQUIRED
Hashes	SHA1, InitiatingProcessSHA1	File collection and prevalence enrichment.	OPTIONAL

Data quality requirements

- Confirm DLL-load telemetry is enabled and retained.
- Validate path equality semantics with raw events.
- Preserve both files and their hashes.
- Inventory legitimate Azure emulator deployments.

Known failure conditions

- Module-load collection is unavailable.
- The actor renames the loader, DLL or directory.
- Paths are absent or normalized incompatibly.
- The activity predates telemetry retention.



DETECTION LOGIC

Primary-platform logic with every condition traceable to evidence.

HUNTING QUERY

DETECTION OBJECTIVE

Find the documented FDMTP side-load relationship.

Select only csmonitor.exe loading the named co-located DLL from a directory ending in quickfox\updated.

ANALYTICAL LOGIC

Direct filters and investigable output.

Four direct predicates; no IOC list, join, sequence, threshold, aggregation, rarity or automatic attribution.

KQL · UNTESTED HUNT

```
// QuickFox / FDMTP - csmonitor.exe DLL sideloading
// Classification: Hunting query - Untested implementation sketch
// Confidence: Medium
// Primary platform: Microsoft Defender XDR
// Telemetry: Microsoft Defender for Endpoint DeviceImageLoadEvents
//
// Source evidence: FortiGuard IR observed csmonitor.exe loading a malicious
// Microsoft.ServiceHosting.Tools.dll from the same
// ...\\quickfox\\updated directory in both loader generations.
//
// A match does not prove FDMTP execution or attribution. Validate the loader
// and DLL hashes, signatures, file origin, software ownership, and follow-on
// process/network activity before disposition.

DeviceImageLoadEvents
| where InitiatingProcessFileName =~ "csmonitor.exe"
| where FileName =~ "Microsoft.ServiceHosting.Tools.dll"
| where FolderPath =~ InitiatingProcessFolderPath
| where FolderPath endswith @"\quickfox\updated"
| project Timestamp,
    DeviceId,
    DeviceName,
    InitiatingProcessAccountName,
    InitiatingProcessFolderPath,
    InitiatingProcessCommandLine,
    InitiatingProcessSHA1,
    FolderPath,
    FileName,
    SHA1,
    ReportId
| order by Timestamp desc
```

REVIEW FIELDS · BASELINE LOCALLY · VALIDATE BEFORE PRODUCTION

STATUS	THRESHOLD	JOIN / SEQUENCE	PRIMARY KEY	DISPOSITION
Untested hunt	None	None	DeviceName + Timestamp + ReportId	Manual triage

What a match means

The endpoint recorded the source-observed loader, module and staging-directory relationship and requires immediate file validation.

What a match does not prove

Malicious DLL contents, successful FDMTP execution, C2 activity, plugin delivery, data theft or attribution.



TRIAGE & INVESTIGATION

Evidence collection before escalation.

SOC WORKFLOW

NUMBERED WORKFLOW

- 01

Confirm the raw event.
Verify loader, module, both paths, hashes, device and event identifiers.
- 02

Validate file trust.
Collect both files and review signatures, hashes, prevalence and acquisition time.
- 03

Establish software origin.
Confirm QuickFox version, installer source and organizational ownership.
- 04

Build the file timeline.
Look for update.zip, extraction activity, update.bin and the 1-byte data.dat artifact.
- 05

Review follow-on activity.
Inspect child processes, network connections and Registry changes after the load.

Escalation gate

Escalate when the pair is newly staged, unowned or hash-matched to the published samples and followed by suspicious network, Registry or plugin activity.

Primary benign overlap

Legitimate Azure emulator installations, packaging workflows, forensic labs and security tests may use csmonitor.exe and the Microsoft.ServiceHosting.Tools.dll name. The exact QuickFox staging suffix materially narrows but does not eliminate overlap.

INVESTIGATION PIVOTS

Host
Coverage, role and neighboring alerts.

Loader
Signature, hash, path and prevalence.

Module
Hash, signer, origin and PE content.

Files
Archive extraction and data.dat creation.

Process
Parent, command line and descendants.

Network
Connections from csmonitor.exe after the module load.

DISPOSITION CRITERIA

OUTCOME	MINIMUM EVIDENCE	ACTION
BENIGN	Approved, owned and repeatable activity.	Document and tune narrowly.
SUSPICIOUS	Unknown activity without corroborated malicious follow-on.	Deepen endpoint review.
LIKELY MALICIOUS	Unauthorized match plus coherent file, process, network or control-plane evidence.	Escalate incident response.



FALSE POSITIVES & VALIDATION

Operational controls required before promotion.

QUALITY GATE

FALSE-POSITIVE MATRIX

SCENARIO	WHY IT OVERLAPS	TUNING ACTION
Azure emulator installation	May legitimately deploy the same executable and library names.	Require approved installation path, publisher, hash and owner.
Software packaging workflow	May stage legitimate components in temporary directories.	Validate package source, ticket, signer and stable path provenance.
Security or malware lab	May reproduce the published loader chain.	Scope to isolated assets and authorized test windows.

Baseline plan

- Run across 30-90 days.
- Inspect every raw match.
- Classify paths, hashes and owners.
- Measure field completeness before alerting.

Validation procedure

- Generate a benign same-directory module load in an isolated host.
- Confirm FolderPath equality and suffix semantics.
- Verify known Azure emulator activity is excluded.
- Review all historical matches and missed variants.

TEST CASES	
CASE	EXPECTED RESULT
Controlled safe pair under quickfox\updated	Expected match with complete paths and hashes.
Same pair in an approved Azure path	No match because the directory suffix differs.
csmonitor.exe loads another DLL	No match.
Another process loads the named DLL	No match.

Hunting	Candidate	Production	Optimized
Initial query	Current status	Measured precision	Stable feedback

Success

Controlled replay matches, approved Azure activity does not, and residual volume is low and explainable.

Rejection

Path semantics cannot establish co-location or legitimate activity frequently reproduces the full relationship.

Failure

DeviceImageLoadEvents is absent or omits required paths.



ATT&CK, CONCLUSION & REFERENCES

Coverage statement, residual risk and document control.

CLOSURE

MITRE ATT&CK MAPPING

TECHNIQUE	NAME	COVERAGE STATEMENT
T1574.001	Hijack Execution Flow: DLL	Directly covers the source-observed side-loading of a malicious library through a legitimate Microsoft executable.

Detection coverage summary

Strong for the exact csmonitor.exe, Microsoft.ServiceHosting.Tools.dll and quickfox\updated relationship; weak for renamed files or alternative staging paths.

Residual risk

The actor can change any campaign artifact, while file content, C2 success and attribution remain outside the event.

ANALYST CONCLUSION

A high-value campaign hunt with narrow, explicit boundaries.

The direct incident-response evidence and documented Defender schema support a selective retrospective hunt. Promotion requires a controlled replay, tenant prevalence measurement and proof that legitimate Azure emulator workflows remain distinguishable.

REFERENCES

- 1. FortiGuard Incident Response Team, "QuickFox Supply Chain Attack Used to Deploy FDMTP Implant," 4 August 2026. <https://www.fortinet.com/blog/threat-research/quickfox-supply-chain-attack-used-to-deploy-fdmtip-implant>
- 2. Microsoft Learn, "DeviceImageLoadEvents table," accessed 5 August 2026. <https://learn.microsoft.com/en-us/defender-xdr/advanced-hunting-deviceimageloadevents-table>
- 3. MITRE ATT&CK, T1574.001. <https://attack.mitre.org/techniques/T1574/001/>

DOCUMENT CONTROL

VERSION	DATE	STATUS	OWNER
v1.0	05 August 2026	Hunting	Detection Engineering

Publication control

Eight A4 pages. URLs are defanged. No customer data, credentials, HTML source or rendering assets are included in the public repository.